

Week 07. Multiple Linear Regression. Practice.

Part 2: Statistics and Econometrics. MK68-2023/24

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Causal interpretation of the MLR

Multiple Regression Controls for the factors that matter...

- These are only the factors that are included in the model.

 All other factors that are not included in the MLR are assumed to remain constant!

- External Ceteris Paribus is assumed.

Violation of the Ceteris Paribus causes Bias.

Regression assumptions

They ensure that linear regression estimates obtained using the OLS method are **unbiased** and **efficient**.

- **OLS is unbiased: Assumptions 1 to 4 are satisfied**
 1. Linearity
 2. Random Sampling
 3. No Collinearity
 4. No Endogeneity
- **OLS is unbiased and efficient or Best Linear Unbiased Estimator (BLUE)**
 5. Homoscedasticity (No Autocorrelation)
- **OLS is a Classical linear model (CLM). Assumptions 1 to 6 are satisfied**
 6. Error Terms Normality

Example 1. Hedonic Land prices model

1. Problem setting

We would like to assess the effect of the “Conservation Reserve Program” (CPR) on agricultural land prices.

Conservation Reserve Program

- is a subsidy
- obligates farms NOT TO GROW ANY CROPS on the enrolled land
- pays monetary compensation in exchange;

This is a riskless income, so it may increase the land price.

- However, if the land is productive, such set-aside measures may reduce farmland prices.

2. Regression model formulation

The hedonic price model ([Palmquist 1989](#)) is an econometric approach to quantifying monetary values of differentiated characteristics of goods or services, which are subjects of economic exchange.

The theory of the hedonic price model allows us to estimate the price as a function of all characteristics of a good. Available variables in our data ([Taff and Weisberg 2007](#)) are:

- **acrePrice** - sale price in dollars per acre;
- **crpPct** - the percentage of the farm area enrolled in CRP;
- **acres** - the size of the farm in acres;
- **region** - a region in the state of Minnesota;
- **tillable** - the percentage of the farm that is rated arable by the assessor;
- **productivity** - average agronomic productivity scaled from 1 to 100, with larger numbers for more productive land;

EQ: Write down a regression equation that could help answering the main research question.

$$\begin{aligned} \text{acrePrice} = & \alpha + \beta_1(\text{crpPct}) + \beta_2(\text{acres}) + \\ & \beta_3(\text{region}_{\text{West Central}}) + \beta_4(\text{region}_{\text{Central}}) + \beta_5(\text{region}_{\text{South West}}) + \\ & \beta_6(\text{region}_{\text{South Central}}) + \beta_7(\text{region}_{\text{South East}}) + \beta_8(\text{tillable}) + \\ & \beta_9(\text{productivity}) + \epsilon \end{aligned}$$

EQ: Formulate the ex-ante hypothesis about the sign and magnitude of the coefficient “_____”. In 1-2 sentences, explain why (based on what theory, common sense) you make this hypothesis.

3 Exploratory data analysis

3.1 The data

acrePrice	acres	region	tillable	crpPct	productivity
2933	303	West Central	75	23	53
2283	265	West Central	75	3	52
2150	40	South West	90	20	72
2432	110	West Central	60	50	48
1494	77	Northwest	90	93	45
2495	192	West Central	97	17	64
4086	220	West Central	95	3	65
1400	78	West Central	56	13	43
2092	75	West Central	100	13	69
1852	58	West Central	93	34	43
1458	120	Northwest	88	69	34
2600	77	West Central	96	4	57
1051	97	South West	82	73	50
2544	117	South West	98	3	59
2013	75	South Central	96	5	42

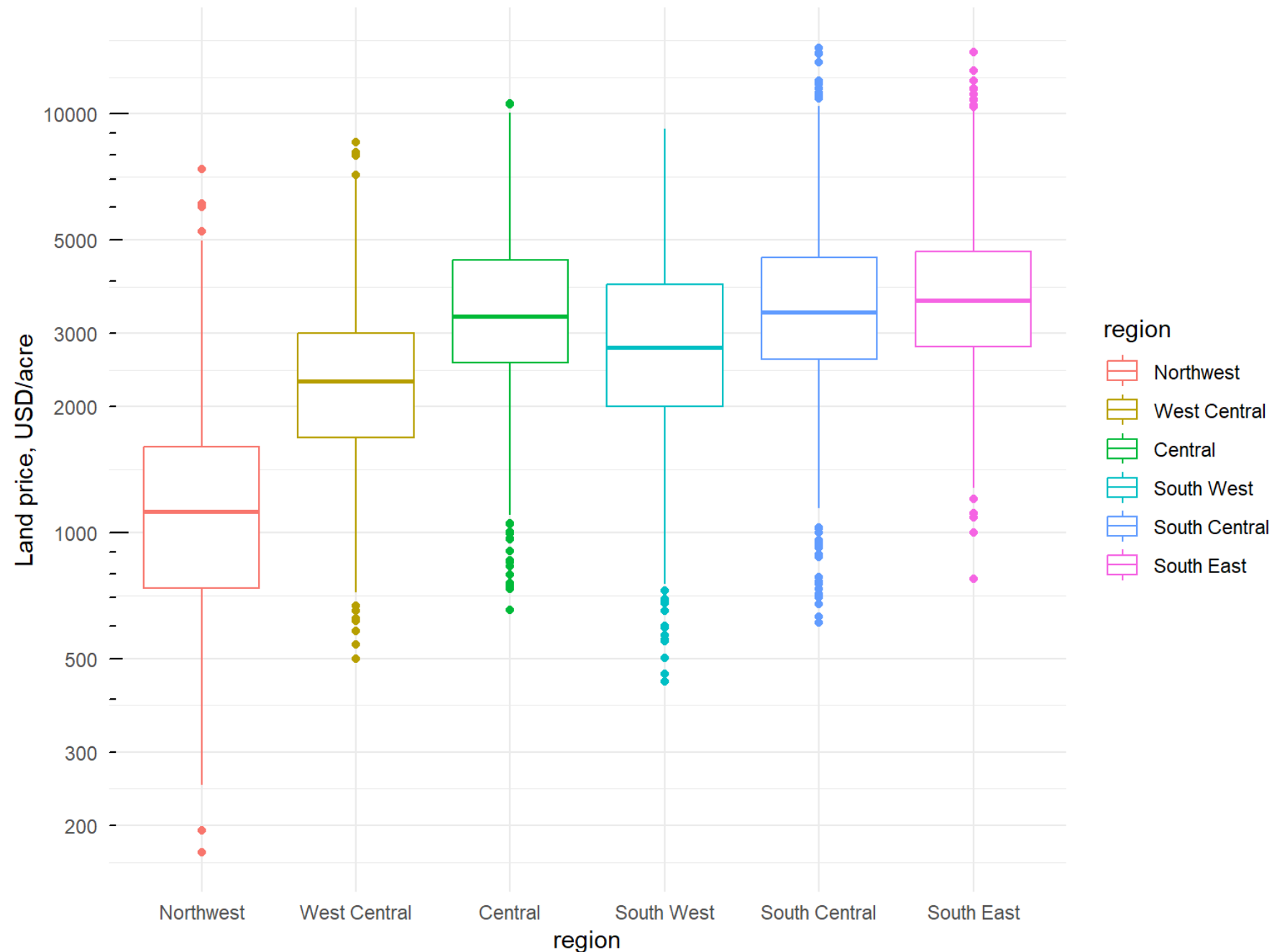
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3.2 Summary statistics

	N	Mean	SD	Min	Median	Max	Histogram
acrePrice	8787	3044.24	1553.36	173.00	2800.00	14379.00	
acres	8787	114.79	88.15	1.00	80.00	1365.00	
tillable	8787	88.19	15.82	5.00	94.00	100.00	
crpPct	8787	2.93	13.63	0.00	0.00	100.00	
productivity	8787	66.74	13.48	1.00	68.00	99.00	

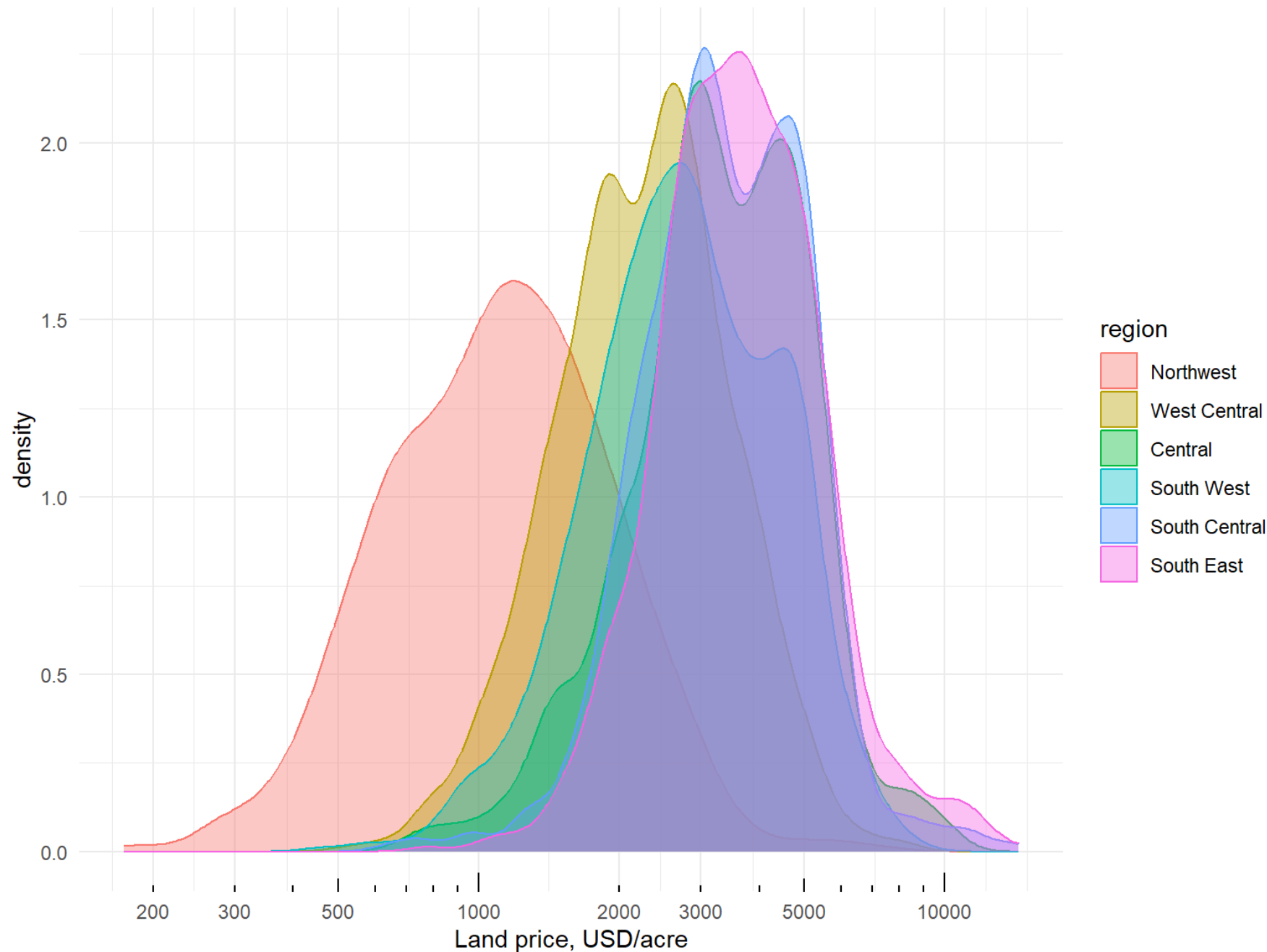
region	N	%
Northwest	788	9.0
West Central	1747	19.9
Central	1058	12.0
South West	2438	27.7
South Central	1983	22.6
South East	773	8.8
	8787	100.0

3.3 Regional differences



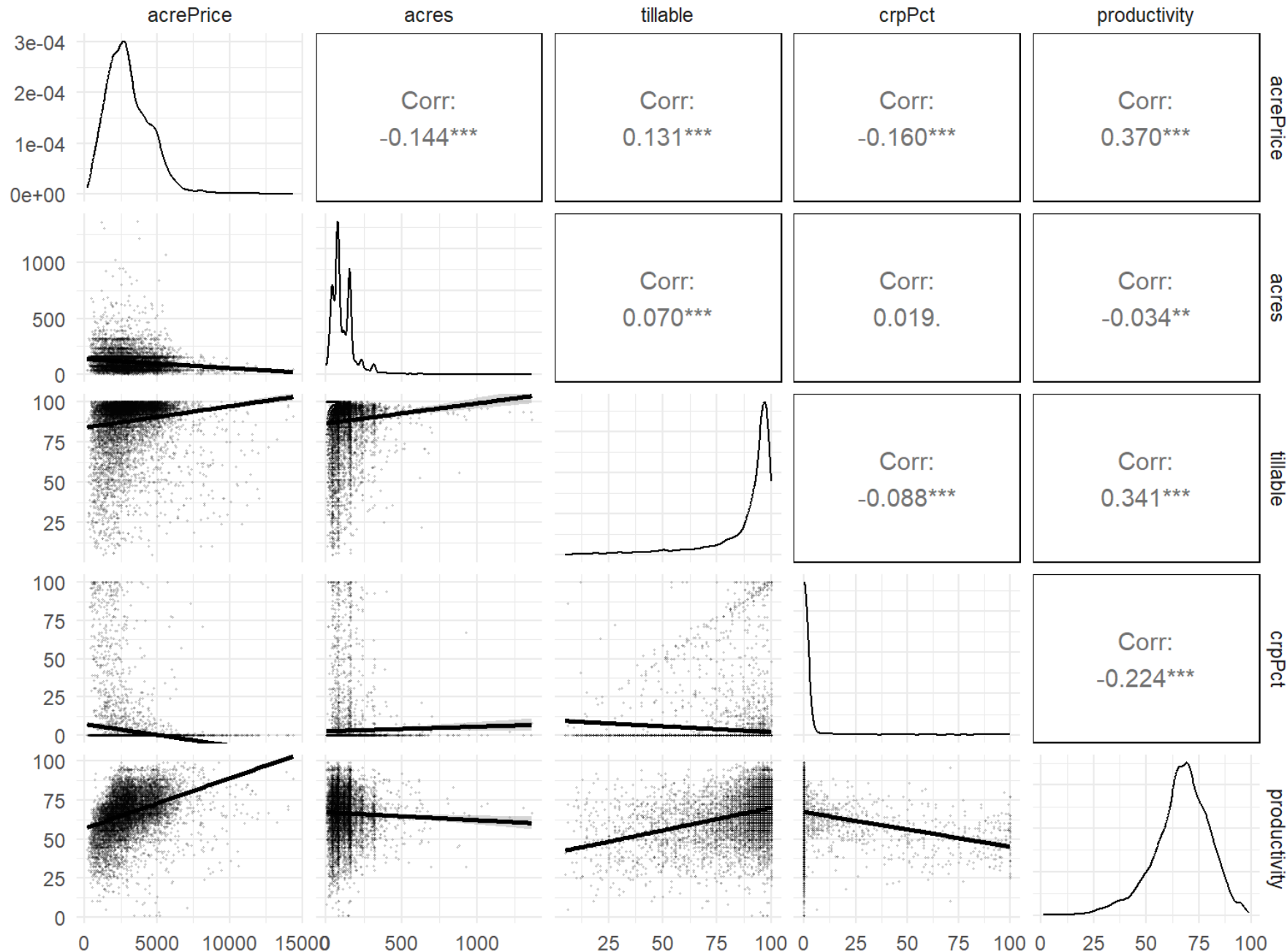
- What kind of plot is this?
- **What (if any) transformation of the Y axis is used on the plot?**
- What can we conclude about the land prices by region?
- **EQ: What are the possible causes of price differences between the regions?**
- **EQ: Does the region cause land prices to be different?**

3.4 Distributions



- What kind of plot is this?
- What does it show?
- **What (if any) transformation of the X axis is used on the plot?**
- What differences and similarities does it manifest?

3.5 Scatter plots



- EQ: What kind of relationship exists between **acrePrices** (land price per acre, USD) and **acres** (plot size in acres) variables? (Linear/non linear, strong/weak correlation, positive/negative)
- EQ: same question for any other pair of variables.

4. Estimating the regression

```

1 fit_land_0 <- lm(
2   acrePrice ~ crpPct + acres + region + tillable + productivity,
3   landprice_dta)
4 summary(fit_land_0)
5 # Coefficients:
6 #               Estimate Std. Error t value Pr(>|t|)
7 # (Intercept)      80.2822   102.3493    0.784   0.4328
8 # crpPct           -6.9429     1.0727   -6.473 1.02e-10 ***
9 # acres            -1.2898     0.1650   -7.818 6.00e-15 ***
10 # regionWest Central  995.2924    58.0391   17.149 < 2e-16 ***
11 # regionCentral    1950.2114    64.5434   30.215 < 2e-16 ***
12 # regionSouth West  1291.8278    58.3426   22.142 < 2e-16 ***
13 # regionSouth Central 1747.5506    62.5597   27.934 < 2e-16 ***
14 # regionSouth East  2040.6768    72.4923   28.150 < 2e-16 ***
15 # tillable          2.1123     0.9691    2.180  0.0293 *
16 # productivity      23.6903     1.3300   17.812 < 2e-16 ***
17 #
18 # Residual standard error: 1332 on 8777 degrees of freedom
19 # Multiple R-squared:  0.2654, Adjusted R-squared:  0.2647
20 # F-statistic: 352.4 on 9 and 8777 DF, p-value: < 2.2e-16

```

6. Results interpretation

Parameter	Coefficient	SE	t(8777)	p
(Intercept)	80.28	102.35	0.78	0.433
crpPct	-6.94	1.07	-6.47	< .001
acres	-1.29	0.16	-7.82	< .001
region (West Central)	995.29	58.04	17.15	< .001
region (Central)	1950.21	64.54	30.22	< .001
region (South West)	1291.83	58.34	22.14	< .001
region (South Central)	1747.55	62.56	27.93	< .001
region (South East)	2040.68	72.49	28.15	< .001
tillable	2.11	0.97	2.18	0.029
productivity	23.69	1.33	17.81	< .001

Model: acrePrice ~ crpPct + acres + region + tillable + productivity (8787 Observations) Residual standard deviation: 1332.012 (df = 8777) R2: 0.265; adjusted R2: 0.265

Note units of measurement:
acrePrice sale price in USD per acre;

crpPct is % of all farm acres enrolled under the program (CRP);

acres size of the farm in acres;

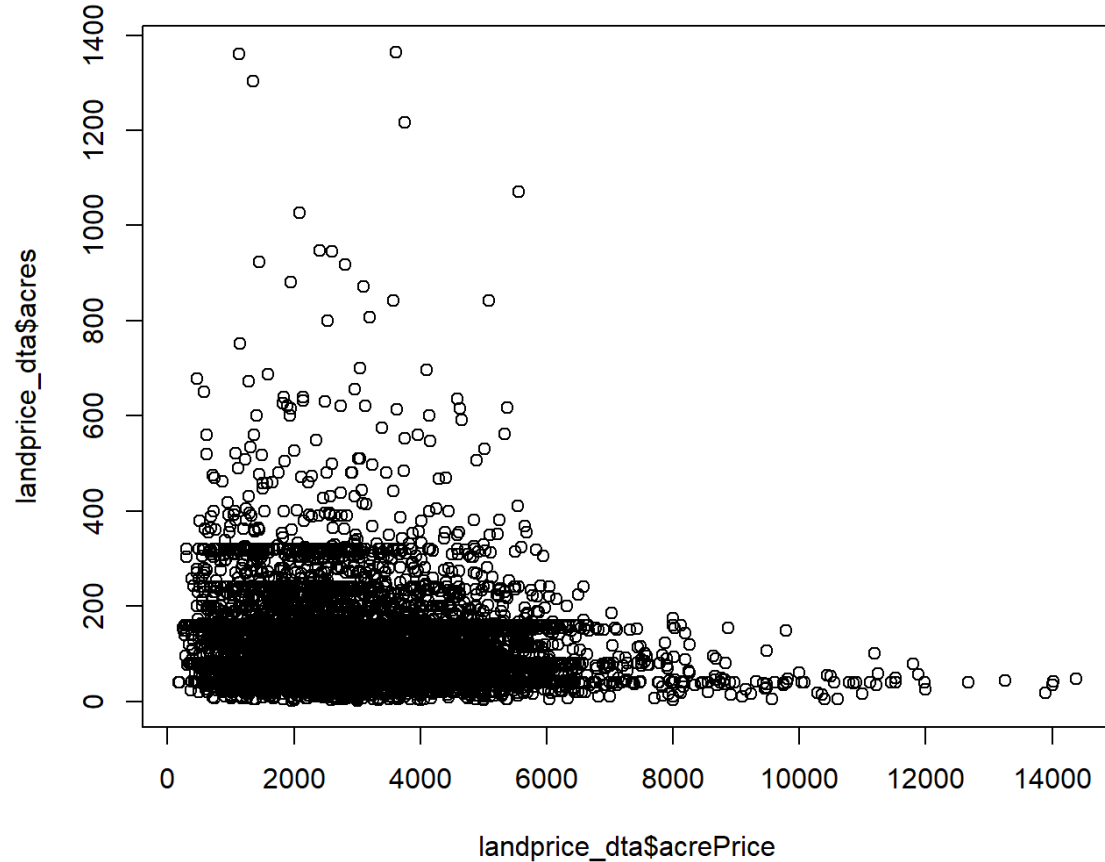
region is the sub-regions in the state of Minnesota;

tillable is the percentage of the farm that is rated arable by the assessor;

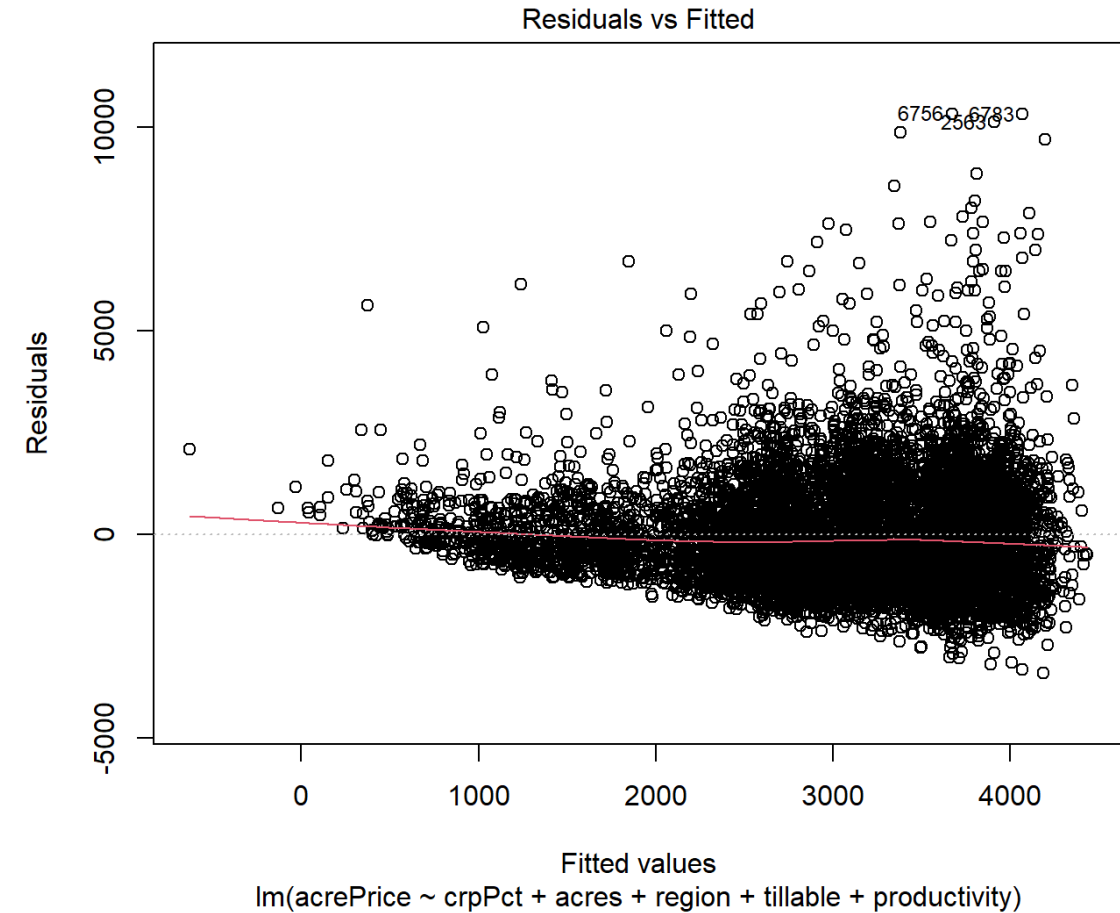
productivity is the average agronomic productivity scaled from 1 to 100, with larger numbers for more productive land.

EQ: Write down the full interpretation of and perform a hypothesis testing on each coefficient.

5. Checking the linearity assumption



Is the relationship between `acrePrice` and `acres` linear?



Is the linearity assumption satisfied in our regression?

4. Regression re-specification

Let us transform **acrePrice** and **acres** variables with the log transformation and estimate the regression again. The old regression is:

$$\begin{aligned} \text{acrePrice} = & \alpha + \beta_1(\text{crpPct}) + \beta_2(\text{acres}) + \\ & \beta_3(\text{region}_{\text{West Central}}) + \beta_4(\text{region}_{\text{Central}}) + \beta_5(\text{region}_{\text{South West}}) + \\ & \beta_6(\text{region}_{\text{South Central}}) + \beta_7(\text{region}_{\text{South East}}) + \beta_8(\text{tillable}) + \\ & \beta_9(\text{productivity}) + \epsilon \end{aligned}$$

Write down the new regression equation with the transformations.

$$\begin{aligned} \log(\text{acrePrice}) = & \alpha + \beta_1(\text{crpPct}) + \beta_2(\log(\text{acres})) + \\ & \beta_3(\text{region}_{\text{West Central}}) + \beta_4(\text{region}_{\text{Central}}) + \beta_5(\text{region}_{\text{South West}}) + \\ & \beta_6(\text{region}_{\text{South Central}}) + \beta_7(\text{region}_{\text{South East}}) + \beta_8(\text{tillable}) + \\ & \beta_9(\text{productivity}) + \epsilon \end{aligned}$$

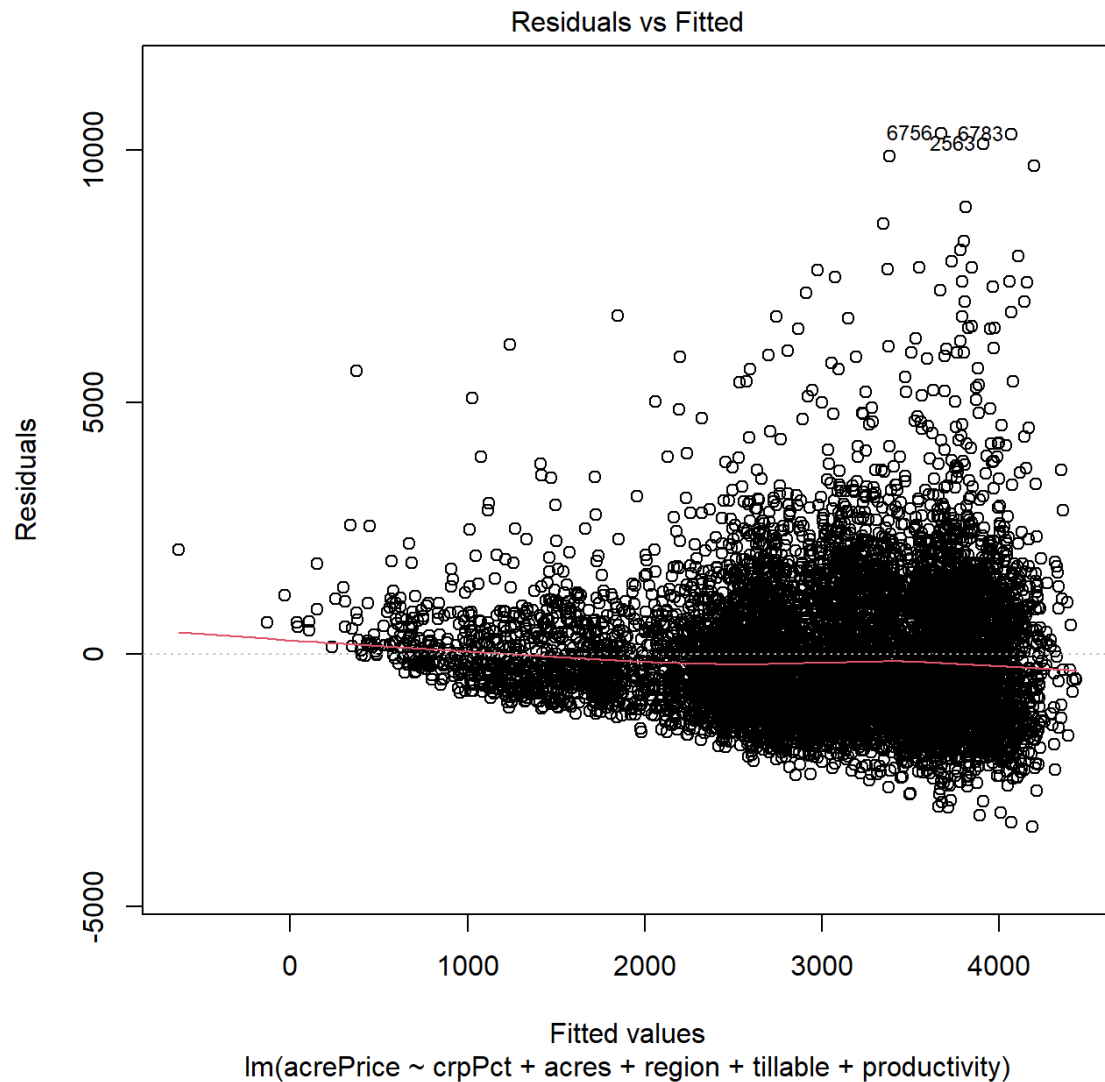
4.1 Results of the re-specified regression

	log(acrePrice)
(Intercept)	6.608 (0.042)***
crpPct	-0.003 (0.000)***
log(acres)	-0.069 (0.007)***
regionWest Central	0.624 (0.018)***
regionCentral	0.936 (0.021)***
regionSouth West	0.712 (0.019)***
regionSouth Central	0.849 (0.020)***
regionSouth East	0.939 (0.023)***
tillable	0.003 (0.000)***
productivity	0.009 (0.000)***
Num.Obs.	8787
R2	0.408
R2 Adj.	0.408

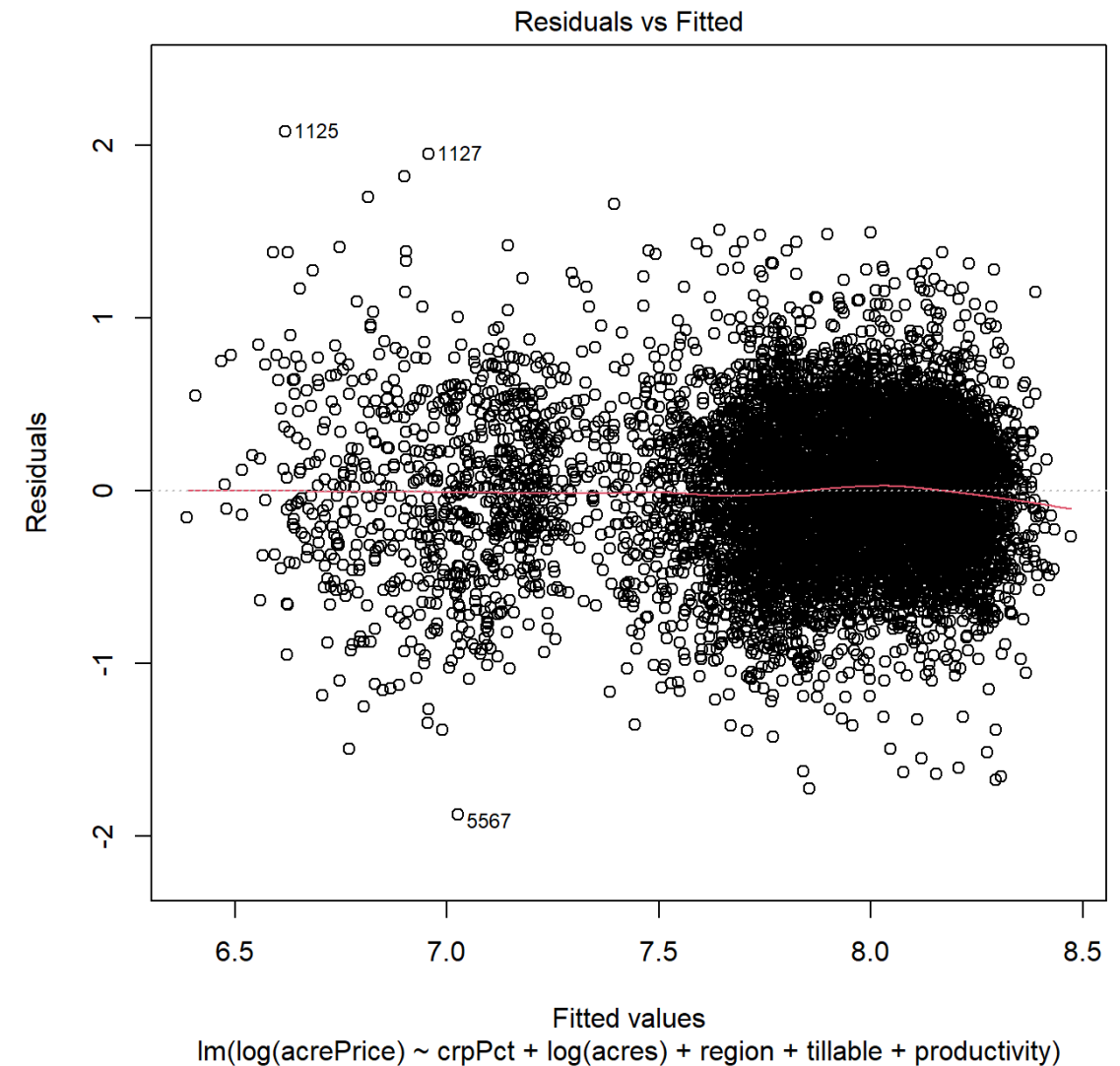
Note: ^^ p-values are: ‘+’ $p < 0.1$, ‘*’ $p < 0.05$, ‘**’ $p < 0.01$, ‘***’ $p < 0.001$. Standard errors are reported in parentheses

5. Re-checking Linearity

Before:



After the log transformation:



6.1 Results interpretation

	log(acrePrice)
(Intercept)	6.608 (0.042)***
crpPct	-0.003 (0.000)***
log(acres)	-0.069 (0.007)***
regionWest Central	0.624 (0.018)***
regionCentral	0.936 (0.021)***
regionSouth West	0.712 (0.019)***
regionSouth Central	0.849 (0.020)***
regionSouth East	0.939 (0.023)***
tillable	0.003 (0.000)***
productivity	0.009 (0.000)***
Num.Obs.	8787
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Note: ^^ p-values are: '+' p < 0.1, '*' p < 0.05, '***' p < 0.01, '****' p < 0.001. Standard errors are reported in parentheses

Log transformation requires adjusting the interpretation:

$\log(y) = \beta \log(x)$ is a **log-log** transformation

- 1 % (**percent**) change in x causes β % (**percent**) change in y (AKA Elasticity).

$\log(y) = \beta x$ is a **log-level** transformation

- 1 **unit** change in x causes 100β % (**percent**) change in y .

$y = \beta \log(x)$ is a **level-log** transformation

- 1 % (**percent**) change in x causes $\beta/100$ **unit** in y .

EQ: Interpret selected coefficients and perform statistical hypothesis testing on them.

8. Comparing two models and concluding

	acrePrice	log(acrePrice)
(Intercept)	80.282 (102.349)	6.608 (0.042)***
crpPct	-6.943 (1.073)***	-0.003 (0.000)***
acres	-1.290 (0.165)***	
regionWest Central	995.292 (58.039)***	0.624 (0.018)***
regionCentral	1950.211 (64.543)***	0.936 (0.021)***
regionSouth West	1291.828 (58.343)***	0.712 (0.019)***
regionSouth Central	1747.551 (62.560)***	0.849 (0.020)***
regionSouth East	2040.677 (72.492)***	0.939 (0.023)***
tillable	2.112 (0.969)*	0.003 (0.000)***
productivity	23.690 (1.330)***	0.009 (0.000)***
log(acres)		-0.069 (0.007)***
Num.Obs.	8787	8787
R2	0.265	0.408
R2 Adj.	0.265	0.408

Note: ^^ p-values are: '+' p < 0.1, '*' p < 0.05, '***' p < 0.01, '****' p < 0.001. Standard errors are reported in parentheses

Conclude:

EQ: Which model would you choose to explain the effect of **crpPct** on the price of land?

EQ: Which model best explains the relationship between X and Y?

What is the effect of **crpPct** on land price?

Which estimated effects are different from your ex-ante hypothesis? Provide a plausible explanation of why are they different?

Example 2. Production function

1. Problem setting

We would like to evaluate the causal effect of the BIMAS production intensification program on the rice output among rice-producing farmers in India.

- The outcome variable is:
 - **goutput** - gross output of rice in kg per year produced on the farm.
- Key variable of interest is:
 - **bimas variable of interest** is an intensification program. Each farmer either participates in it (yes), partially participate (mixed), or does not participate (no).
- Other control variables are:
 - **size** the total area cultivated with rice, measured in hectares;
 - **urea** urea application in kilogram
 - **phosphate** phosphate application in kilogram
 - **hiredlabor** hired labor in hours
 - **famlabor** family labor in hours

2. Regression model

The production function is used to estimate the relationship. We consider two main production functions (see [Feng and Horrace 2012](#)):

1. Linear [production function](#), when output is a linear combination of all inputs. **(Write it down)**

$$\begin{aligned} \text{goutput} = & \alpha + \beta_1(\text{size}) + \beta_2(\text{bimas}_{\text{yes}}) + \\ & \beta_3(\text{bimas}_{\text{mixed}}) + \beta_4(\text{famlabor}) + \beta_5(\text{hiredlabor}) + \\ & \beta_6(\text{urea}) + \beta_7(\text{phosphate}) + \epsilon \end{aligned}$$

2. [Cobb-Douglas](#) production function, when the log of output is a linear combination of logged inputs. **(Write it down)**

$$\begin{aligned} \log(\text{goutput}) = & \alpha + \beta_1(\log(\text{size})) + \beta_2(\text{bimas}_{\text{yes}}) + \\ & \beta_3(\text{bimas}_{\text{mixed}}) + \beta_4(\log(\text{famlabor})) + \beta_5(\log(\text{hiredlabor})) + \\ & \beta_6(\log(\text{urea})) + \beta_7(\log(\text{phosphate})) + \epsilon \end{aligned}$$

2.1 Ex-ante hypotheses about the coefficients

$$\begin{aligned} \text{goutput} = & \alpha + \beta_1(\text{size}) + \beta_2(\text{bimas}_{\text{yes}}) + \\ & \beta_3(\text{bimas}_{\text{mixed}}) + \beta_4(\text{famlabor}) + \beta_5(\text{hiredlabor}) + \\ & \beta_6(\text{urea}) + \beta_7(\text{phosphate}) + \epsilon \end{aligned}$$

EQ: Formulate the ex-ante hypotheses about coefficients?

3 Exploratory data analysis

3.1 The data

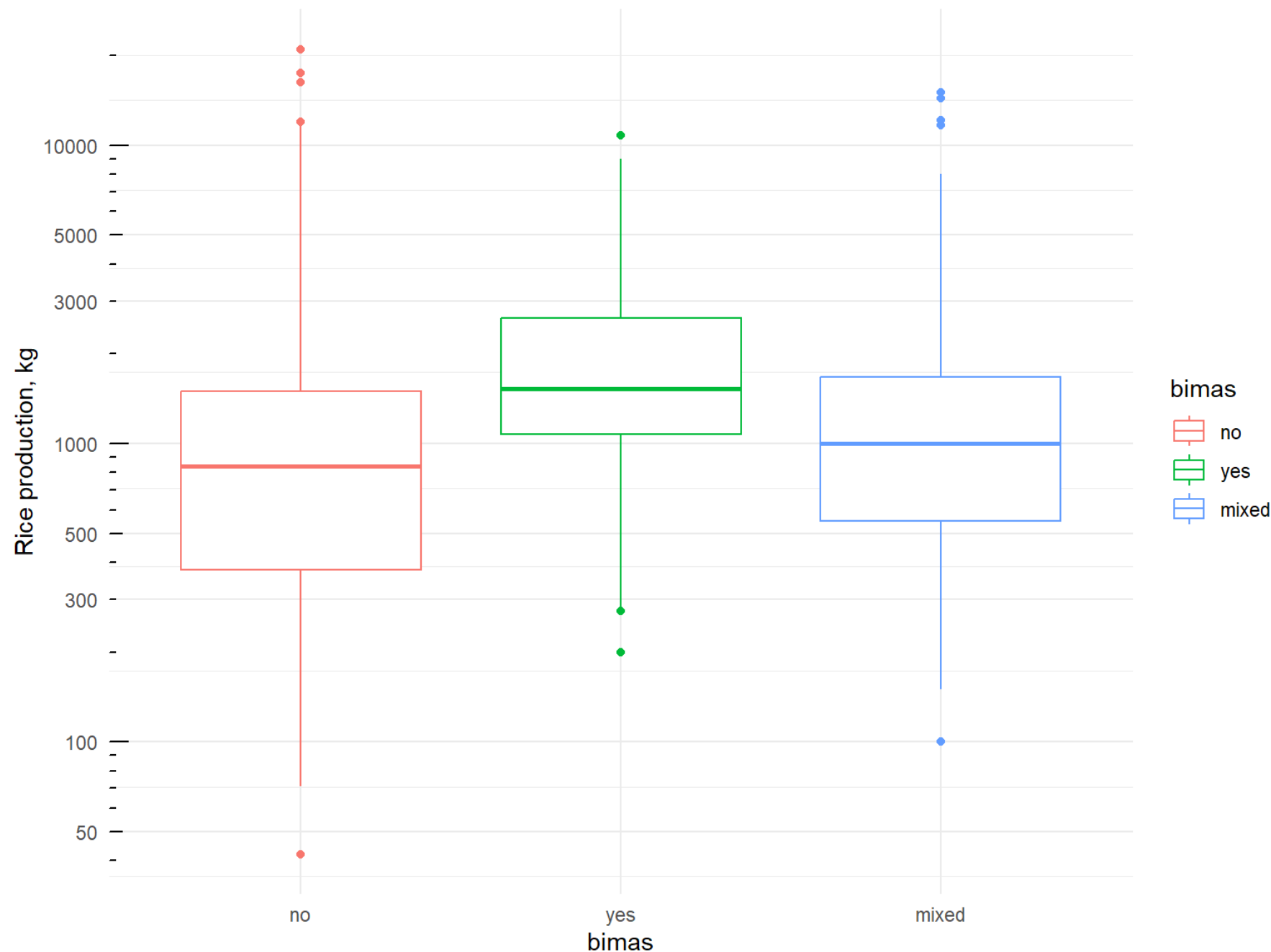
goutput	size	bimas	famlabor	hiredlabor	urea	phosphate
71	0.020	no	25	1	5	2
886	0.282	no	66	365	30	10
6010	2.898	mixed	134	1878	300	50
952	0.215	no	233	60	50	20
640	0.143	no	68	56	51	25
280	0.143	no	76	42	32	10
960	0.357	no	75	305	50	10
240	0.184	no	71	40	40	10
1830	0.575	yes	148	395	165	60
1106	0.192	no	141	39	28	57
4400	0.857	mixed	130	1015	200	50
436	0.143	mixed	33	98	50	50
450	0.140	mixed	89	39	25	50
2700	0.400	mixed	284	220	150	75
1041	0.351	no	48	160	85	15

3.2 Summary statistics

	N	Mean	SD	Min	Median	Max	Histogram
goutput	883	1449.76	1956.91	42.00	952.00	20960.00	
size	883	0.43	0.52	0.01	0.29	5.32	
famlabor	883	152.76	147.63	1.00	113.00	1526.00	
hiredlabor	883	243.47	422.93	1.00	121.00	4536.00	
urea	883	99.80	128.00	1.00	65.00	1250.00	
phosphate	883	39.19	49.17	1.00	25.00	700.00	

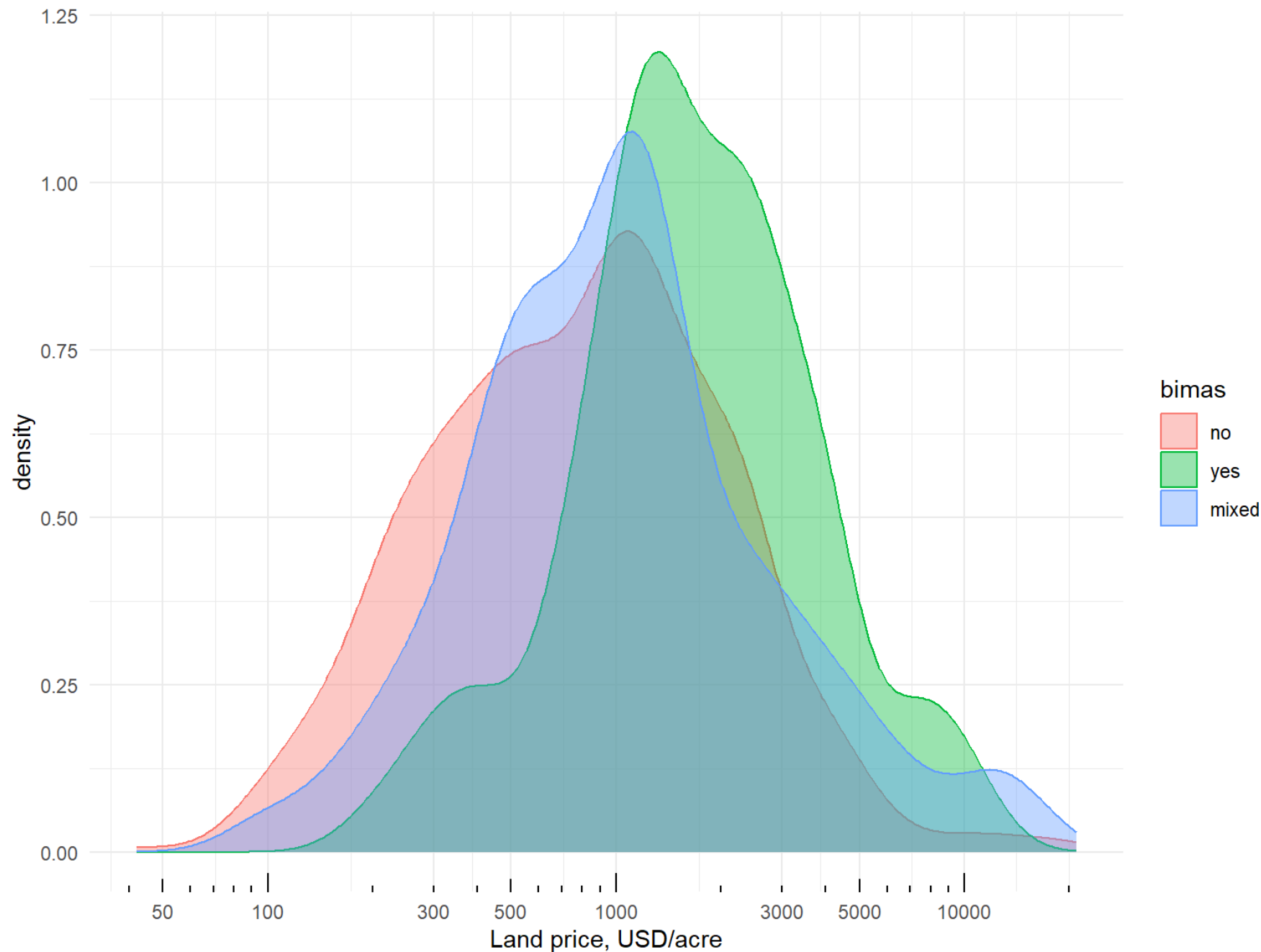
bimas	N	%
no	661	74.9
yes	84	9.5
mixed	138	15.6

3.3 What effect BIMAS program have?



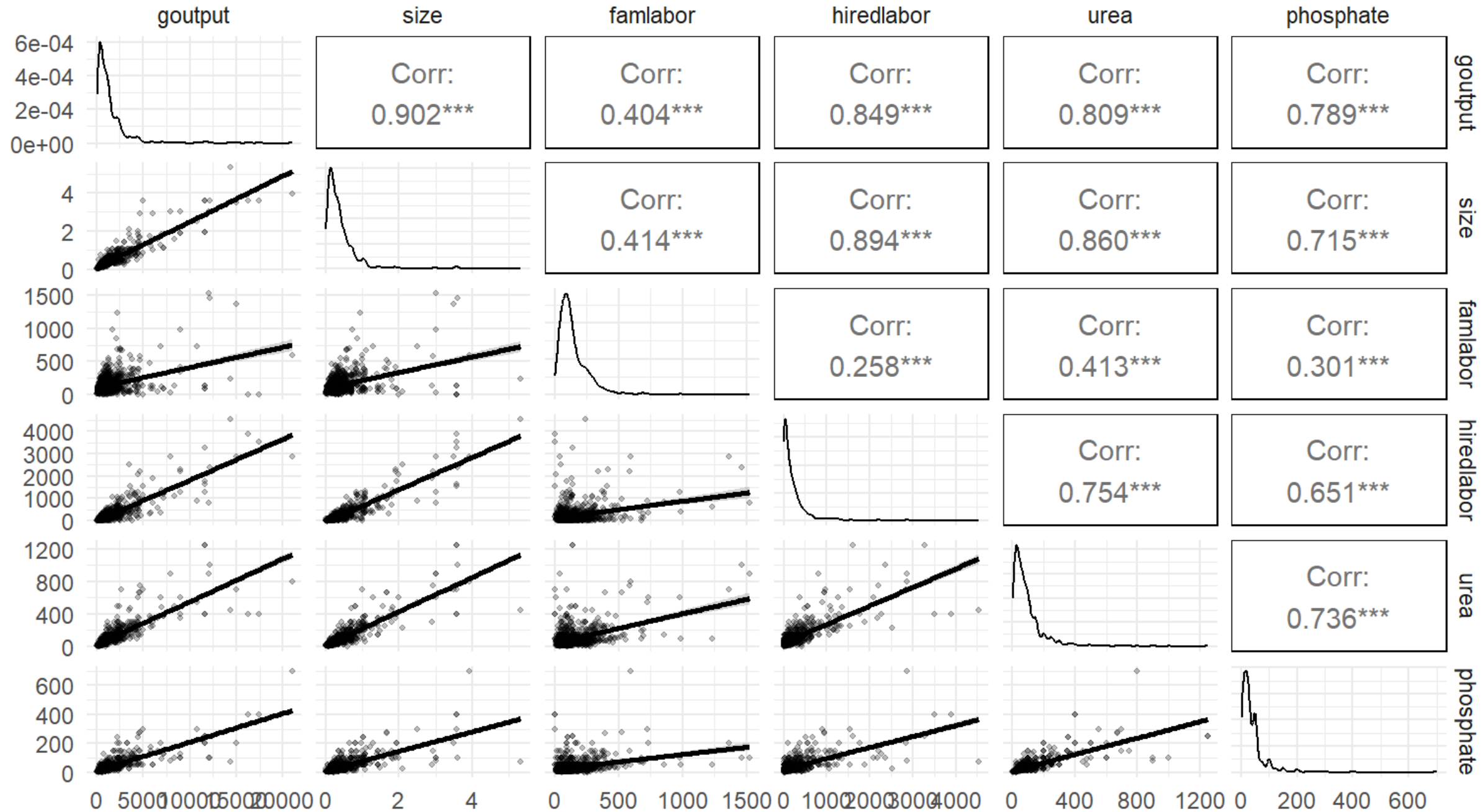
- What kind of plot is this?
- What can we conclude about the differences in rice output by program participation?
- EQ: Does the program participation cause rice output to be different?

3.4 Distributions



- What kind of plot is this?
- What does it show?
- What differences and similarities does it manifest?

3.5 Scatter plots



4. Estimating a regression

```

1 fit_rice_0 <- lm(
2   goutput ~ size + bimas + famlabor + hiredlabor + urea + phosphate,
3   rice_dta_0)
4 summary(fit_rice_0)
5 # Coefficients:
6 #               Estimate Std. Error t value Pr(>|t|)
7 # (Intercept) -107.91531   38.36178  -2.813 0.005017 **
8 # size        1824.02155  138.40695  13.179 < 2e-16 ***
9 # bimasyes     157.87341   83.12605   1.899 0.057866 .
10 # bimasmixed -262.07581   68.13525  -3.846 0.000129 ***
11 # famlabor     0.78902    0.18633   4.234 2.53e-05 ***
12 # hiredlabor   1.03722    0.13132   7.898 8.47e-15 ***
13 # urea        -0.03429    0.39242  -0.087 0.930381
14 # phosphate   11.08610    0.74446  14.891 < 2e-16 ***
15 #
16 # Residual standard error: 706.6 on 875 degrees of freedom
17 # Multiple R-squared:  0.8707, Adjusted R-squared:  0.8696
18 # F-statistic: 841.5 on 7 and 875 DF, p-value: < 2.2e-16

```

6. Interpretation (0)

$$\widehat{\text{goutput}} = -107.92 + 1824.02(\text{size}) + 157.87(\text{bimas}_{\text{yes}}) - 262.08(\text{bimas}_{\text{mixed}}) + 0.79(\text{famlabor}) + 1.04(\text{hiredlabor}) - 0.03(\text{urea}) + 11.09(\text{phosphate})$$

This first regression is in **level-level** form (without any linear transformation). Thus, interpretations are straightforward:

- 1 unit change in x causes $\hat{\beta}$ unit change in y .

Linear prod. fun (level-level)

Dep. var.	Rice output, kg
(Intercept)	-107.915 (38.362)**
Size, HA	1824.022 (138.407)***
BIMAS [YES]	157.873 (83.126)+
BIMAS [MIXED]	-262.076 (68.135)***
Family labor, h	0.789 (0.186)***
Hired labor, h	1.037 (0.131)***
Urea, kg	-0.034 (0.392)
Phosphate, kg	11.086 (0.744)***
Num.Obs.	883
R2	0.871
R2 Adj.	0.870

Note: ^^ p-values are ‘+’ $p < 0.1$, ‘*’ $p < 0.05$, ‘***’ $p < 0.01$, ‘****’ $p < 0.001$. Standard errors are reported in parentheses

6. Interpretation (1)

size is the total area cultivated with rice, measured in hectares.

- An increase in the farm size by 1 hectare, causes on average the gross output of rice to increase by 1824 kg, holding all other factors constant.
- The effect of farm size is significant at the 0.1% level of significance (H_0 is rejected based on the p-value, which is $p < 0.001$ given “***”) suggesting that a positive relationship between the farm size and output is likely to be present in the population.

Linear prod. fun (level-level)	
Dep. var.	Rice output, kg
(Intercept)	-107.915 (38.362)**
Size, HA	1824.022 (138.407)***
BIMAS [YES]	157.873 (83.126)+
BIMAS [MIXED]	-262.076 (68.135)***
Family labor, h	0.789 (0.186)***
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Note: ^^ p-values are ‘+’ $p < 0.1$, ‘*’ $p < 0.05$, ‘**’ $p < 0.01$, ‘***’ $p < 0.001$. Standard errors are reported in parentheses

6. Interpretation (2)

i **bimas** is an intensification program. Each farmer either:

- participate in it [YES], or partially participate [MIXED] or do not participate [NO].
 - Because the dummy variable for [NO] is omitted, it is implied in the intercept.
 - Variables [YES] and [MIXED] show the effect of different levels of participation **compared to NO (non-participation in BIMAS)**.
- Participating in the BIMAS on a MIXED basis reduces the gross output of rice by 262 kg compared to non-participation given that all other factors remain fixed. The effect is significant at the 5% level of significance.

Linear prod. fun (level-level)

Dep. var.	Rice output, kg
(Intercept)	-107.915 (38.362)**
Size, HA	1824.022 (138.407)***
BIMAS [YES]	157.873 (83.126)+
BIMAS [MIXED]	-262.076 (68.135)***
Family labor, h	0.789 (0.186)***
Hired labor, h	1.037 (0.131)***
Urea, kg	-0.034 (0.392)
Phosphate, kg	11.086 (0.744)***
Num.Obs.	883
R2	0.871
R2 Adj.	0.870

Note: ^^ p-values are ‘+’ $p < 0.1$, ‘*’ $p < 0.05$, ‘***’ $p < 0.01$, ‘****’ $p < 0.001$. Standard errors are reported in parentheses

6. Interpretation (3)



- Fully participating in the BIMAS program [YES] yields an estimate of the coefficient of 157.8 implying on average a 158 kg increase in rice production; however, this result is insignificant at the 5% level.
- The present functional form fails to reject H_0 about the significant difference of this estimate from zero concluding that such an effect may result from the random variation in the sample, rather than exist in the population.
- To further investigate this effect, we might consider modifying the functional form or collecting more data.

Linear prod. fun (level-level)

Dep. var.	Rice output, kg
(Intercept)	-107.915 (38.362)**
Size, HA	1824.022 (138.407)***
BIMAS [YES]	157.873 (83.126)+
BIMAS [MIXED]	-262.076 (68.135)***
Family labor, h	0.789 (0.186)***
Hired labor, h	1.037 (0.131)***
Urea, kg	-0.034 (0.392)
Phosphate, kg	11.086 (0.744)***
Num.Obs.	883
R2	0.871
R2 Adj.	0.870

Note: ^^ p-values are '+', p < 0.1, '*' p < 0.05, '**' p < 0.01, '***' p < 0.001. Standard errors are reported in parentheses

6. Interpretation (4) DIY



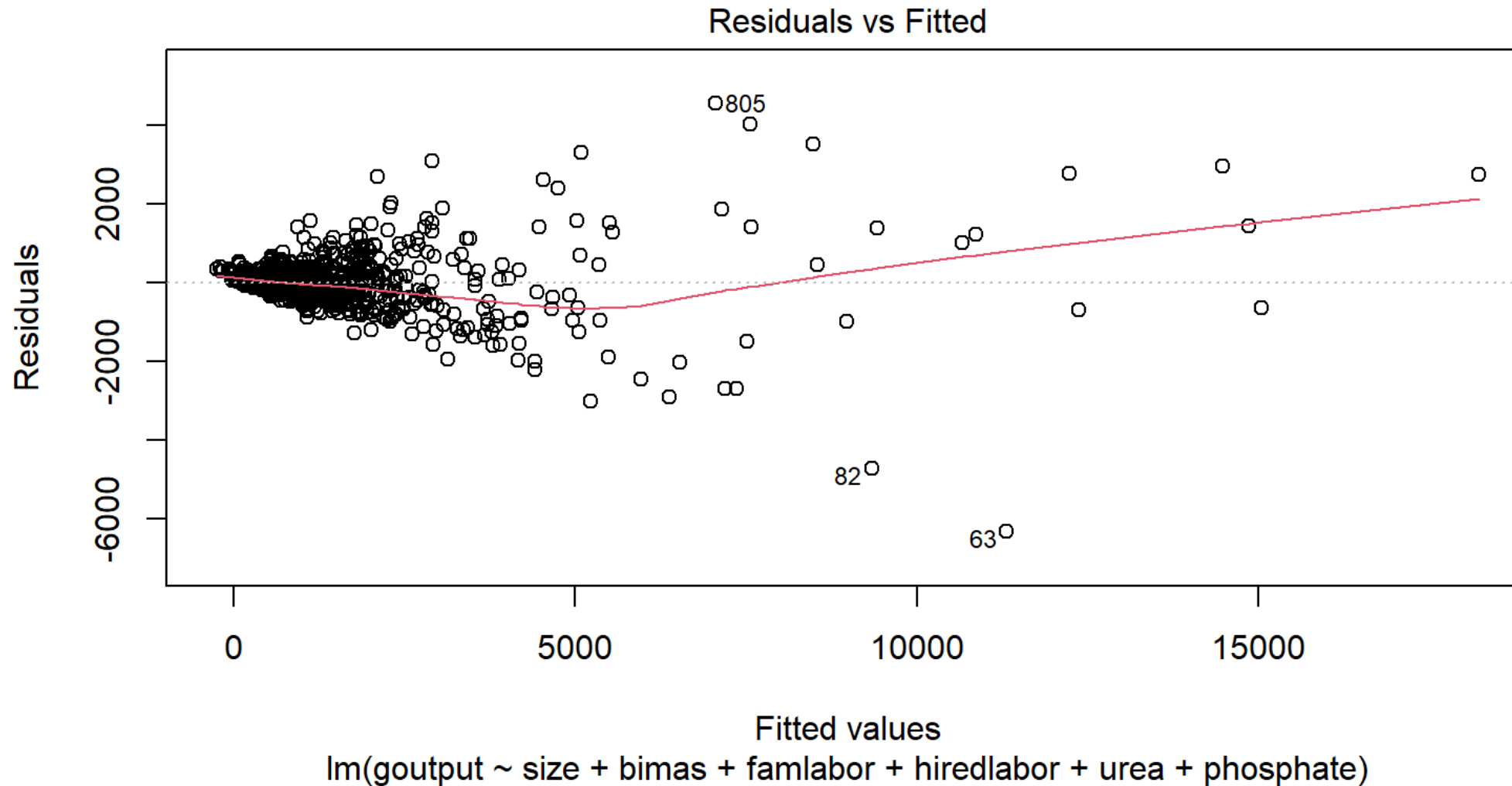
- `urea` or `phosphate` application in kilogram;
- `hiredlabor` or `famlabor` in hours;

Linear prod. fun (level-level)

Dep. var.	Rice output, kg
(Intercept)	-107.915 (38.362)**
Size, HA	1824.022 (138.407)***
BIMAS [YES]	157.873 (83.126)+
BIMAS [MIXED]	-262.076 (68.135)***
Family labor, h	0.789 (0.186)***
Hired labor, h	1.037 (0.131)***
Urea, kg	-0.034 (0.392)
Phosphate, kg	11.086 (0.744)***
Num.Obs.	883
R2	0.871
R2 Adj.	0.870

Note: ^^ p-values are ‘+’ $p < 0.1$, ‘*’ $p < 0.05$, ‘***’ $p < 0.01$, ‘****’ $p < 0.001$. Standard errors are reported in parentheses

5. Checking the linearity assumption (1)



- Clear signs of non-linearity.

4. Model re-specification

Estimating the Cobb-Douglas production function. Note, that $\log(\cdot)$ stands for a natural logarithm.

$$\begin{aligned} \log(\widehat{\text{goutput}}) = & 5.72 + 0.55(\log(\text{size})) + 0.12(\text{bimas}_{\text{yes}}) - \\ & 0.14(\text{bimas}_{\text{mixed}}) + 0.04(\log(\text{famlabor})) + 0.03(\log(\text{hiredlabor})) + \\ & 0.25(\log(\text{urea})) + 0.13(\log(\text{phosphate})) \end{aligned}$$

4.1 Comparing regressions

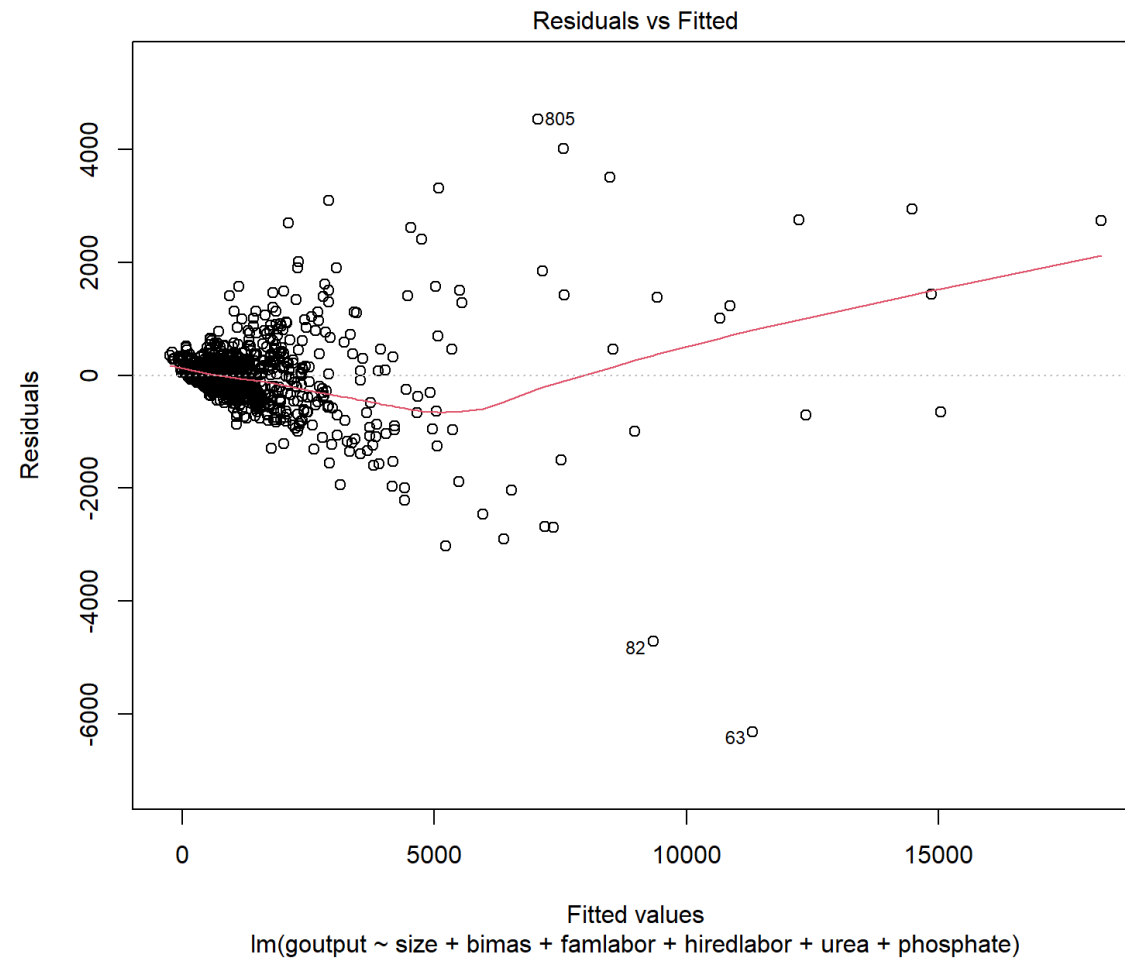
	Linear (level-level)	Cobb-Douglas(log-log)
Dep. var.	Rice output, kg	log(Rice output, kg)
BIMAS [YES]	157.873 (83.126)+	0.118 (0.041)**
BIMAS [MIXED]	-262.076 (68.135)***	-0.142 (0.033)***
Size, HA	1824.022 (138.407)***	
log(Size, HA)		0.555 (0.027)***
Family labor, h	0.789 (0.186)***	0.042 (0.015)**
Hired labor, h	1.037 (0.131)***	0.035 (0.011)**
Urea, kg	-0.034 (0.392)	0.247 (0.028)***
Phosphate, kg	11.086 (0.744)***	0.133 (0.021)***
(Intercept)	-107.915 (38.362)**	5.721 (0.150)***
Num.Obs.	883	883
R2	0.871	0.878
R2 Adj.	0.870	0.877

- What has changed?
- Has any coefficient changed the sign?
- Has any coefficient become significant / insignificant?
- How have the goodness of fit measures (GOF) changed?

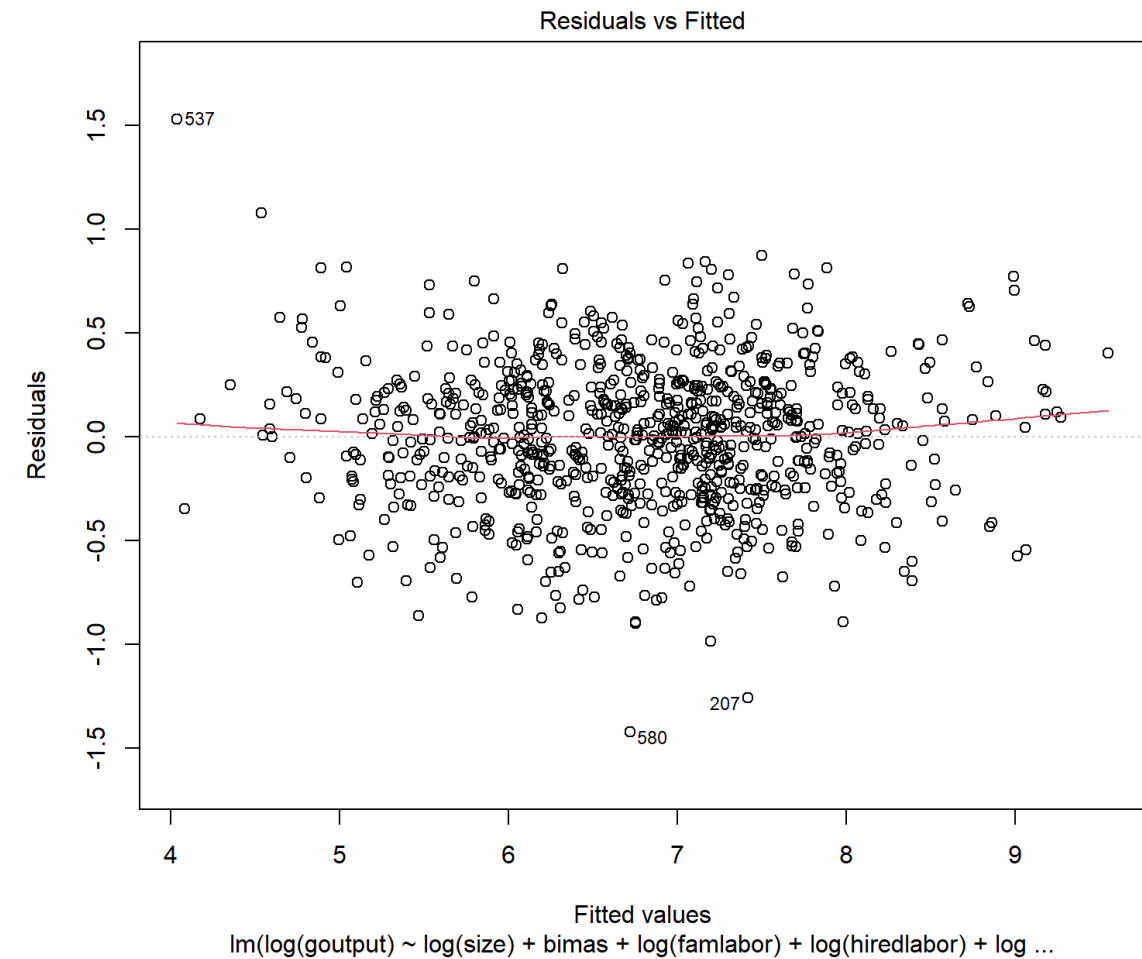
Note: ^^ p-values are ‘+’ $p < 0.1$, ‘*’ $p < 0.05$, ‘***’ $p < 0.01$, ‘****’ $p < 0.001$. Standard errors are reported in parentheses

5. (Again) Checking Linearity

Linear production function



Cobb-Douglas production function



Substantial improvement in linearity.

6. Interpretation with log (0)

This regression is in the **log-log** transformation for all variables except for the dummy variables, which are in the **log-level**.

log-log interpretation is straightforward:

- 1 % percent increase in x causes $\hat{\beta}$ % (percent) change in y .

Cobb-Douglas (log-log)	
Dep. var.	log(Rice output, kg)
BIMAS [YES]	0.118 (0.041)**
BIMAS [MIXED]	-0.142 (0.033)***
log(Size, HA)	0.555 (0.027)***
Family labor, h	0.042 (0.015)**
Hired labor, h	0.035 (0.011)**
Urea, kg	0.247 (0.028)***
Phosphate, kg	0.133 (0.021)***
(Intercept)	5.721 (0.150)***
Num.Obs.	883
R2	0.878
R2 Adj.	0.877

Note: ^^ p-values are ‘+’ $p < 0.1$, ‘*’ $p < 0.05$, ‘***’ $p < 0.01$, ‘****’ $p < 0.001$. Standard errors are reported in parentheses

6. Interpretation with log (1)

log-level interpretation of the dummy variables is more complex:

- For the purpose of the exam, please be able to explain whether the dummy variable has a positive or negative effect on y and if this effect is significant.

For your information and future, please be informed that:

- Change (increase) of a dummy variable x from 0 to 1 causes $100 \cdot (e^{\hat{\beta}} - 1)$ % change in y .

Cobb-Douglas (log-log)	
Dep. var.	log(Rice output, kg)
BIMAS [YES]	0.118 (0.041)**
BIMAS [MIXED]	-0.142 (0.033)***
log(Size, HA)	0.555 (0.027)***
Family labor, h	0.042 (0.015)**
Hired labor, h	0.035 (0.011)**
Urea, kg	0.247 (0.028)***
Phosphate, kg	0.133 (0.021)***
(Intercept)	5.721 (0.150)***
Num.Obs.	883
R2	0.878
R2 Adj.	0.877

Note: ^^ p-values are '+', $p < 0.1$, '*' $p < 0.05$, '**' $p < 0.01$, '***' $p < 0.001$. Standard errors are reported in parentheses

6. Interpretation with log (2)

$\log(\text{size})$ the total area cultivated with rice, measured in hectares;

- $\log\text{-}\log$ transformation.
- Increase of the farm size by 1% is associated with the 0.56% increase in the rice output, given that all other factors remain fixed.
- These results are significant at the 5% significance level.

Cobb-Douglas (log-log)	
Dep. var.	log(Rice output, kg)
BIMAS [YES]	0.118 (0.041)**
BIMAS [MIXED]	-0.142 (0.033)***
log(Size, HA)	0.555 (0.027)***
Family labor, h	0.042 (0.015)**
Hired labor, h	0.035 (0.011)**
Urea, kg	0.247 (0.028)***
Phosphate, kg	0.133 (0.021)***
(Intercept)	5.721 (0.150)***
Num.Obs.	883
R ²	0.878
R ² Adj.	0.877

Note: ^^ p-values are '+', p < 0.1, '*' p < 0.05, '***' p < 0.01, '****' p < 0.001. Standard errors are reported in parentheses

6. Interpretation with log (3)

- Participating in the BIMAS on a MIXED basis reduces the gross output of rice by $100 \cdot e^{-0.142} - 1 = -100 \cdot 0.132 = -13.2$ % (reduces by 13.2%) compared to non-participation given that all other factors remain fixed. The effect is significant at the 5% level of significance.
- Participating in the BIMAS on a full basis (yes) basis increases the gross output of rice by: $100 \cdot e^{0.118} - 1 = 100 \cdot 0.125 = 12.5$ % (increases by 12.2%) compared to non-participation given that all other factors remain fixed. The effect is significant at the 5% level of significance.

Cobb-Douglas (log-log)	
Dep. var.	log(Rice output, kg)
BIMAS [YES]	0.118 (0.041)**
BIMAS [MIXED]	-0.142 (0.033)***
log(Size, HA)	0.555 (0.027)***
Family labor, h	0.042 (0.015)**
Hired labor, h	0.035 (0.011)**
Urea, kg	0.247 (0.028)***
Phosphate, kg	0.133 (0.021)***
(Intercept)	5.721 (0.150)***
Num.Obs.	883
R2	0.878
R2 Adj.	0.877

Note: ^^ p-values are ‘+’ $p < 0.1$, ‘*’ $p < 0.05$, ‘***’ $p < 0.01$, ‘****’ $p < 0.001$. Standard errors are reported in parentheses

6. Interpretation with log (4) DIY

- **urea** or **phosphate** application in kilogram;
- **hiredlabor** or **famlabor** in hours

Cobb-Douglas (log-log)	
Dep. var.	log(Rice output, kg)
BIMAS [YES]	0.118 (0.041)**
BIMAS [MIXED]	-0.142 (0.033)***
log(Size, HA)	0.555 (0.027)***
Family labor, h	0.042 (0.015)**
Hired labor, h	0.035 (0.011)**
Urea, kg	0.247 (0.028)***
Phosphate, kg	0.133 (0.021)***
(Intercept)	5.721 (0.150)***
Num.Obs.	883
R2	0.878
R2 Adj.	0.877

Note: ^^ p-values are ‘+’ $p < 0.1$, ‘*’ $p < 0.05$, ‘***’ $p < 0.01$, ‘****’ $p < 0.001$. Standard errors are reported in parentheses

8. Conclusions

What is the causal effect of BIMAS program on the gross output of rice?

What is the goodness of fit of the model?

EQ: Which model explains the relationship between inputs and outputs more accurately (with less bias)?

EQ: Which model has a higher explanatory power?

	Linear (level-level)	Cobb-Douglas(log-log)
Dep. var.	Rice output, kg	log(Rice output, kg)
BIMAS [YES]	157.873 (83.126)+	0.118 (0.041)**
BIMAS [MIXED]	-262.076 (68.135)***	-0.142 (0.033)***
Size, HA	1824.022 (138.407)***	
log(Size, HA)		0.555 (0.027)***
Family labor, h	0.789 (0.186)***	0.042 (0.015)**
Hired labor, h	1.037 (0.131)***	0.035 (0.011)**
Urea, kg	-0.034 (0.392)	0.247 (0.028)***
Phosphate, kg	11.086 (0.744)***	0.133 (0.021)***
(Intercept)	-107.915 (38.362)**	5.721 (0.150)***
Num.Obs.	883	883
R2	0.871	0.878
R2 Adj.	0.870	0.877

Note: ^^ p-values are '+', p < 0.1, '**' p < 0.05, '***' p < 0.01, '****' p < 0.001. Standard errors are reported in parentheses

Example 3. Duflo, Kremer, and Robinson (2008) How High Are Rates of Return to Fertilizer? Evidence from Field Experiments in Kenya

Objective

In (Duflo, Kremer, and Robinson 2008) authors are trying to estimate the effect of fertilizers application on farms profit.

They employ and randomized controlled trial experiment randomly selecting farmers between children's parents in certain schools.

On each farm, they demarcate 3 experimental fields and train farmers how to apply certain quantities of fertilizers on these fields. They also supervised the farmers over the growing seasons accurately collected the harvest and measured the outcome each time.

In each trial, random plots (out of the demarcated ones) were selected on each farm for applying different levels of fertilizers.

The authors measured:

- returns as $[\text{output value of (treatment - control)} - \text{inputs cost}] / [\text{inputs cost}]$
- yields change $(\text{treatment} - \text{control}) / (\text{control}) * 100$

Results

TABLE 1—RETURNS TO FERTILIZER

	Mean (1)	Median (2)	Std. error (3)	Obs. (4)
<i>Panel A. 1/4 Teaspoon Top Dressing Fertilizer</i>				
Percentage increase in yield	28.1	8.9	6.8	112
Rate of return over the season	4.8	−27.7	38.8	112
Annualized rate of return (at the mean and median)	8.4	−42.6		112
<i>Panel B. 1/2 Teaspoon Top Dressing Fertilizer</i>				
Percentage increase in yield	47.6	24.3	6.1	200
Rate of return over the season	36.0	23.9	16.9	202
Annualized rate of return (at the mean and median)	69.5	44.4		202
<i>Panel C. 1 Teaspoon Top Dressing Fertilizer</i>				
Percentage increase in yield	63.1	30.6	8.2	273
Rate of return over the season	−10.8	−16.9	8.4	274
Annualized rate of return (at the mean and median)	−17.8	−27.3		274

Column (1) reports the mean in returns and yields between control and treatment fields for different rates of fertilizer application overall growing seasons.

- What yield change is associated with different levels of fertilizer application?
- Can we conclude that fertilizer application improves yields/returns on average in the population?
- What rates of fertilizer application improve yields/returns on average in the population?

Is the ceteris paribus ensured in this RCT?

Authors were criticized if the ceteris paribus holds in their experiment

- specifically concerning the monetary return levels.
- monetary outcomes often depend on many observed and unobserved factors, therefore, they are often seen as “endogenous” in econometrics.
 - What other factors can affect the return?
 - How can we ensure that other factors do not affect the return?

Authors solution is the regression analysis

$$\begin{aligned}\text{Return} = & \beta_0 + \beta_1 \text{ Farm size} + \beta_2 \text{ Education} + \beta_3 \text{ Income} \\ & + \beta_4 \text{ Experience in fertilizer use} + \dots \\ & + \beta_k \text{ Other factors} + \epsilon\end{aligned}$$

If other factors affect the return,

- What signs/values coefficients β . should have?
- Formulate ex-ante hypotheses about the expected signs of the coefficients.

Checks and balances (1)

TABLE 2—CROSS-SECTIONAL RELATIONSHIP BETWEEN RETURNS TO TOP DRESSING AND BASE RETURNS

	(1)	(2)	(3)	(4)
Weight of maize on control plot			−0.028 (0.039)	−0.188 (0.124)
Indicator for long rains season			−2.479 (1.506)	
Education	0.019 (0.071)	−0.275 (0.370)	0.01 (0.071)	
Income in past month (in 1,000 Kenyan shillings)	0.000 (0.066)	0.102 (0.261)	0.002 (0.066)	
Household had ever used fertilizer before	0.578 (0.545)	5.984 (3.020)*	0.591 (0.543)	
House has mud walls	0.353 (0.802)	4.400 (4.097)	0.308 (0.801)	
Acres of land owned	−0.005 (0.045)	0.027 (0.170)	−0.010 (0.045)	
Rate of return in previous demonstration		−0.312 (0.820)		
School controls	YES	YES	YES	NO
Individual fixed effects	NO	NO	NO	YES
Observations	323	59	323	122
p-value for joint significance of school controls	0.16	0.04	0.10	
R-squared	0.08	0.23	0.09	0.04

Note: p-values are ‘*’ $p < 0.1$, ‘**’ $p < 0.05$, ‘***’ $p < 0.01$. Standard errors are reported in parentheses.

Checks and balances (2)

Regression in Excel

You can use Excel to perform a regression analysis. There are many tutorials online.

See for example: <https://youtu.be/9wX1a1J4WOI>

References

- Duflo, Esther, Michael Kremer, and Jonathan Robinson. 2008. “How High Are Rates of Return to Fertilizer? Evidence from Field Experiments in Kenya.” *American Economic Review* 98 (2): 482–88.
<https://doi.org/10.1257/aer.98.2.482>.
- Feng, Qu, and William C Horrace. 2012. “Alternative Technical Efficiency Measures: Skew, Bias and Scale.” *Journal of Applied Econometrics* 27 (2): 253–68.
- Palmquist, Raymond B. 1989. “Land as a Differentiated Factor of Production: A Hedonic Model and Its Implications for Welfare Measurement.” *Land Economics* 65 (1): 23. <https://doi.org/10.2307/3146260>.
- Taff, Steven J, and Sanford Weisberg. 2007. “Compensated Short-Term Conservation Restrictions May Reduce Sale Prices.” *Appraisal Journal* 75 (1).

